

Literature Status: Isomorphic Invariance and Reflexivity for the Weak Maximizing Property

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Status: literature already answered. Questions 3.2 of arXiv:1906.02162 ask whether the weak maximizing property (WMP) is invariant under isomorphism and whether WMP is equivalent to reflexivity. The separate later paper arXiv:2103.10288 explicitly answers both questions negatively in Corollary 2.3.

1 Original questions

For Banach spaces E, F , the pair (E, F) has the WMP if every bounded operator $T : E \rightarrow F$ possessing a non-weakly-null maximizing sequence attains its norm. Aron, García, Pellegrino, and Teixeira ask on arXiv PDF page 9 [1]:

Source Questions 3.2. *Are there infinite-dimensional spaces E, F_1, F_2 such that (E, F_1) has the WMP but (E, F_2) does not? Is WMP preserved when the domain or range is replaced by an isomorphic copy? Is a Banach space reflexive exactly when it has the WMP?*

The source had already proved that WMP implies reflexivity, so the final question asks whether the converse holds.

2 Separate later answer

Dantas, Jung, and Martínez-Cervantes state in the introduction of arXiv:2103.10288 that they answer Questions 3.2. Their Corollary 2.3, on arXiv PDF pages 4–5, says:

Later answer (Corollary 2.3). *There are reflexive Banach spaces E_1, E_2, E_3 , all isomorphic to ℓ_2 , such that (ℓ_2, E_1) , (E_2, ℓ_2) , and E_3 fail the WMP.*

The proof applies their Theorem 2.1 to a non-norm-attaining operator on ℓ_2 . Concretely, it uses the renormed finite sums $\ell_2 \oplus_q \mathbb{R}$ for $q < 2$ and $\ell_2 \oplus_p \mathbb{R}$ for $p > 2$, including $p = \infty$.

Since (ℓ_2, ℓ_2) has the WMP, while E_1 is isomorphic to ℓ_2 and (ℓ_2, E_1) fails it, WMP is not preserved by an isomorphic change of the range. Taking $E = \ell_2$, $F_1 = \ell_2$, and $F_2 = E_1$ answers the first existence question. The space E_3 is reflexive but fails WMP, so reflexivity does not imply WMP. The examples involving E_2 also show failure under an isomorphic change of the domain.

Thus every component of Questions 3.2 receives a negative answer. The later paper explicitly identifies these consequences; this packet records a literature match and claims no new theorem.

3 Scope

The separate question in the source about WMP for (L_p, L_q) is not answered in full by this identification. The later paper proves failure when $p > 2$ or $q < 2$ and leaves the standard-norm range $1 < p \leq 2 \leq q < \infty$ open.

References

- [1] R. M. Aron, D. García, D. Pellegrino, and E. V. Teixeira, *Reflexivity and non-weakly null maximizing sequences*, Proc. Amer. Math. Soc. **148** (2020), 741–750; arXiv:1906.02162.
- [2] S. Dantas, M. Jung, and G. Martínez-Cervantes, *Some remarks on the weak maximizing property*, arXiv:2103.10288.